2.3.2 $n \times m = 0 \leftrightarrow n = 0 \vee m = 0$ (Lemma 2.3.3)

$n = 0 \rightarrow n \times m = 0$: $0 \times m = 0$ by definition

$m = 0 \rightarrow n \times m = 0$: base case $n = 0 \rightarrow 0 \times 0 = 0$

Inductively assuming we've defined how $n \times 0 = 0$, for $n++$ we have

$n++ \times 0 = (n \times 0) + 0 = 0 + 0 = 0$

$n \times m = 0 \rightarrow n = 0 \vee m = 0$: by contradiction:

Let's assume there were $n > 0$ and $m > 0$ such that $n \times m = 0$.

Then $\exists k, h$ such that $k++ = n$, $h++ = m$

So $k++ \times h++ = (k \times h++) + h++ = (k \times m) + m$

For the expression above to equal zero, m would have to be zero since it is an added term, a contradiction against the rules of addition